

Spark solution update

A layout-independent mathematical algorithm

Derived from `hy_rk_updateSoln.F90` and its calls

Purpose. Advance a supplied Runge–Kutta stage using already computed face fluxes and gravitational source terms. The following equations specify the cell-local mathematical dependencies without prescribing array layout, scratch vectors, or execution order across cells. They complement `face_flux_solver_algorithm.pdf`.

Scope. The reference is `source/physics/Hydro/HydroMain/Spark/hy_rk_updateSoln.F90`, including its active `update_solution` and `gravSources` calls, and the concrete grid abundance routine in `source/Grid/GridMain/Grid_renormAbundance.F90`. The latter has a no-operation stub in `source/Grid/`; the build-selected implementation determines whether its arithmetic runs. Grid metadata access supplies geometric inputs. Non-Cartesian geometry is rejected by the current update entry point, on both host and device. The commented geometric-source routine is not part of the active algorithm.

Inputs, notation, and result

For each cell C , distinguish two *input* states:

$$V_C^0 = (\rho^0, \mathbf{u}^0, q^0, X_a^0, \mathbf{B}^0, \psi^0, \phi^0), \quad V_C^s = (\rho^s, \mathbf{u}^s, q^s, X_a^s, \mathbf{B}^s, \psi^s, \phi^s). \quad (1)$$

The cell label is suppressed on components when unambiguous. V^0 is the saved reference state; V^s is the stage state before this update. The output V^+ replaces the stage state. q is the stored specific nonmagnetic total energy (`ENER_VAR`); e is the separately stored specific internal energy (`EINT_VAR`). Here q includes kinetic and internal energy but excludes magnetic energy. The conservative energy used for the MHD update includes magnetic energy.

Further inputs are dimension D , positive Cartesian spacings Δx_d , time steps $\Delta t, \Delta t_{\text{old}}$, stage coefficients (a, b, c) , cell gravity $\mathbf{g}_C$, and each directional face flux $F_{C,d}^-, F_{C,d}^+$. These are the negative and positive faces of the *same* cell, both oriented in the positive coordinate direction. Fluxes include species and mass-scalar partial densities. Magnetic quantities are included only for GLM-MHD; potential terms only when enabled.

The output contains updated density, all three velocities, specific energy q , internal energy e , species and mass scalars, and optional magnetic field, magnetic pressure, cleaning field, and potential. This routine does not recompute pressure, temperature, or EOS coefficients.

Step 1. Establish the stage scale and conservative inputs

The three coefficients are arbitrary supplied inputs:

$$a = \text{coeffs}(1), \quad b = \text{coeffs}(2), \quad c = \text{coeffs}(3), \quad \theta = c\Delta t. \quad (2)$$

Do not infer a particular RK method or impose $a + b = 1$ in this routine. Define

$$h_{\min} = \min_{d=1,\dots,D} \Delta x_d, \quad \chi = \frac{h_{\min}}{c\Delta t}. \quad (3)$$

The reference computes χ (`dhdtd`) unconditionally; it is used only for GLM damping. Its literal evaluation presumes $c\Delta t \neq 0$.

For each input $r \in \{0, s\}$ form the conservative components

$$(U^r)_\rho = \rho^r, \quad (4)$$

$$\mathbf{m}^r = \rho^r \mathbf{u}^r, \quad (5)$$

$$E^r = \rho^r q^r + \frac{1}{2} |\mathbf{B}^r|^2, \quad (6)$$

$$Z_a^r = \rho^r X_a^r. \quad (7)$$

Omit the magnetic-energy term for hydrodynamics. In MHD also copy

$$(U^r)_{B_j} = B_j^r \quad (j = x, y, z), \quad (U^r)_\psi = \psi^r. \quad (8)$$

Thus

$$U^r = (\rho^r, \mathbf{m}^r, E^r, Z_1^r, \dots, Z_N^r) \quad (9)$$

with magnetic and cleaning components added for MHD. The original stored internal energy e^r is *not* used to form E^r in this update; the routine uses the stored q^r directly. No ψ energy term is included.

Step 2. Evaluate sources from the pre-update stage

The geometry factors and source in the active implementation are

$$f_- = f_+ = 1, \quad S_{\text{geo}} = 0. \quad (10)$$

With gravity enabled, evaluate

$$(S_C)_\rho = 0, \quad (S_C)_{m_j} = \rho^s g_{j,C}, \quad (11)$$

$$(S_C)_E = \mathbf{m}^s \cdot \mathbf{g}_C, \quad (S_C)_{Z_a} = 0, \quad (12)$$

$$(S_C)_{B_j} = 0, \quad (S_C)_\psi = 0. \quad (13)$$

With gravity disabled, $S_C = 0$ for every component. Gravity uses the stage density and momentum *before* the update, not the saved reference state or the updated/floored state. The force is an acceleration times density; the energy source is momentum dotted with acceleration. Potential extrapolation below does not determine $\mathbf{g}_C$ within this routine.

Step 3. Take the flux divergence and combine RK states

For any conservative component w , define

$$(\mathcal{D}F_C)_w = \sum_{d=1}^D \frac{(F_{C,d}^+)_w - (F_{C,d}^-)_w}{\Delta x_d}. \quad (14)$$

There is no contribution from inactive coordinate directions. Each supplied flux is per unit face area; Cartesian area-to-volume ratios reduce to $1/\Delta x_d$. No flux reconstruction, Riemann solve, or artificial viscosity is performed here.

The provisional conservative update is

$$\boxed{\overline{U}_C = aU_C^0 + bU_C^s + \theta(-\mathcal{D}F_C + S_C)}. \quad (15)$$

This one expression applies to every conservative component, including species partial densities, magnetic fields, and ψ where present. For clarity its main components are

$$\overline{\rho} = a\rho^0 + b\rho^s - \theta(\mathcal{D}F_C)_\rho, \quad (16)$$

$$\overline{m}_j = am_j^0 + bm_j^s - \theta(\mathcal{D}F_C)_{m_j} + \theta\rho^s g_{j,C}, \quad (17)$$

$$\overline{E} = aE^0 + bE^s - \theta(\mathcal{D}F_C)_E + \theta\mathbf{m}^s \cdot \mathbf{g}_C, \quad (18)$$

$$\overline{Z}_a = a\rho^0 X_a^0 + b\rho^s X_a^s - \theta(\mathcal{D}F_C)_{Z_a}, \quad (19)$$

$$\overline{B}_j = aB_j^0 + bB_j^s - \theta(\mathcal{D}F_C)_{B_j}, \quad (20)$$

$$\overline{\psi} = a\psi^0 + b\psi^s - \theta(\mathcal{D}F_C)_\psi. \quad (21)$$

Omit gravity terms when disabled and the last two equations for hydrodynamics. All right-hand inputs must be captured before replacing the stage state. There is no gravitational source in mass, partial density, magnetic, or cleaning-field updates.

Conservation before repair. The same face flux enters its two adjacent cells with opposite divergence signs. For uniform Cartesian spacing, the sum of flux contributions to cell-volume-weighted conserved quantities therefore telescopes to boundary fluxes. The RK blend and sources remain as shown. The subsequent floors and species repair may alter the resulting conserved quantities; they are part of the output algorithm rather than additional face fluxes.

Step 4. Apply GLM damping and remove magnetic energy

For GLM-MHD, store the updated magnetic field directly:

$$\mathbf{B}^+ = \overline{\mathbf{B}}. \quad (22)$$

Damp the provisional cleaning field using configured α and c_h :

$$\boxed{\psi^+ = \overline{\psi} \exp\left(-\frac{\alpha c_h}{\chi}\right) = \overline{\psi} \exp\left(-\frac{\alpha c_h \theta}{h_{\min}}\right).} \quad (23)$$

The second form eliminates the intermediate χ mathematically when $\theta \neq 0$; the source evaluates the first form. The damping is applied *after* the RK blend and flux update and uses $c\Delta t$, not Δt alone. It does not change any other conservative component.

Compute the magnetic-energy density and stored magnetic pressure:

$$M^+ = \frac{1}{2}|\mathbf{B}^+|^2, \quad p_B^+ = M^+. \quad (24)$$

The updated nonmagnetic energy density used to recover stored energy is

$$E_{\text{nm}}^+ = \overline{E} - M^+. \quad (25)$$

For hydrodynamics set $M^+ = 0$ and $E_{\text{nm}}^+ = \overline{E}$, and omit magnetic pressure and cleaning output assignments. No energy correction accounts for ψ damping, because the conservative energy contains no ψ energy.

Step 5. Recover density, velocity, and specific energies

With configured density floor $\rho_{\min}$ and specific-energy floor $e_{\min}$, apply these equations in order:

$$\rho^+ = \max(\overline{\rho}, \rho_{\min}), \quad (26)$$

$$\mathbf{u}^+ = \frac{\overline{\mathbf{m}}}{\rho^+}, \quad (27)$$

$$q^+ = \max\left(e_{\min}, \frac{E_{\text{nm}}^+}{\rho^+}\right), \quad (28)$$

$$k^+ = \frac{1}{2}|\mathbf{u}^+|^2, \quad (29)$$

$$e^+ = \max(e_{\min}, q^+ - k^+). \quad (30)$$

The same energy floor is used for q^+ and e^+ . Density flooring changes the denominator used for velocity, energy, and fractions. The provisional momentum is retained when recovering velocity.

The internal-energy floor does not subsequently reset q^+ to $e^+ + k^+$. Thus stored q^+ and $e^+ + k^+$ can differ when the internal-energy floor is active. No pressure floor or EOS call is performed at this point, and pressure and thermodynamic diagnostics not assigned by the routine retain their prior stage values.

Step 6. Recover mass fractions and extrapolate potential

For every species and every other mass scalar first store

$$\tilde{X}_a^+ = \frac{\bar{Z}_a}{\rho^+}. \quad (31)$$

This uses the floored density from Step 5. It does not use a separately renormalized sum of partial densities. Non-species mass scalars receive no further clipping or normalization in the called grid routine.

When gravitational potential storage is enabled, denote by ϕ^{old} the previous potential supplied in the saved state's `GPOL_VAR`, by ϕ^0 its `GPOT_VAR`, and by ϕ^s the pre-update stage potential. Then

$$\phi^+ = a\phi^0 + b\phi^s + \theta \frac{\phi^0 - \phi^{\text{old}}}{\Delta t_{\text{old}}}. \quad (32)$$

This is an extrapolation using a supplied temporal difference, not a Poisson solve or a potential inferred from newly updated density. The previous-potential field itself is not modified by this routine. The source has no guard for $\Delta t_{\text{old}} = 0$ in this branch. When potential storage is disabled this step is omitted.

Step 7. Apply the selected grid species postprocessing

If species or other mass scalars are present, the host entry point calls `Grid_renormAbundance`. Its concrete `GridMain` implementation acts on species only. Let x_g be its grid abundance floor, which is distinct from the face-reconstruction species floor. For each cell in the supplied normalization region compute

$$Y_a = \max(x_g, \min(1, \tilde{X}_a^+)), \quad a = 1, \dots, N_{\text{sp}}, \quad (33)$$

$$T = \sum_{a=1}^{N_{\text{sp}}} Y_a, \quad \delta = |T - 1|. \quad (34)$$

The concrete routine prints diagnostics if $\delta \geq 0.10$ and aborts if $\delta > 0.10$. If it continues, its final assignments are

$$X_a^+ = \max\left(x_g, \min\left(1, \frac{Y_a}{T}\right)\right). \quad (35)$$

This is *clip*, *sum*, *normalize*, *clip again*. The sum is computed after the first clip and there is no second normalization after the final clip; the final fractions therefore need not sum to exactly one if that clip is active. These assignments change fractions without adjusting density, energy, or fluxes.

The source passes the entire block guard-cell limits to this call, not just the requested update limits. Cells outside the updated region can therefore have their existing species modified by postprocessing; use those cells' existing fractions in the same formulas. If the build selects the stub, no postprocessing occurs. With `OMP_OL` and any species or mass scalars present, the current entry point aborts before this call, after the cell-update loop.

Dependencies and implementation choices

The computation can be expressed as the following cell-local sequence:

1. Read both saved and pre-update stage states and all incident directional face fluxes.
2. Form their conserved quantities; evaluate gravity from the pre-update stage density and momentum.
3. Apply the supplied RK combination and directional flux differences to all conserved components.
4. For MHD, damp ψ , store the updated magnetic field, and subtract its magnetic energy to obtain stored nonmagnetic energy.
5. Floor density, recover velocity, floor specific energy, and recover/floor internal energy; divide partial densities by that same floored density.
6. Extrapolate the potential using saved inputs if enabled.
7. Apply the selected grid abundance procedure over its designated region after the cell updates.

The mathematical vectors $U^0, U^s, \bar{U}, S$ need not be materialized. Each component can be computed in registers, expressions can be fused, and cells can be visited independently. Preserve the pre-update stage momentum until gravitational work is evaluated, and preserve the pre-update stage potential until its extrapolation is evaluated. After the conservative update, only the updated magnetic field and floored density are cross-component dependencies for recovery.

No neighbor *cell states* are needed once face fluxes and gravity are supplied. Each cell needs its $2D$ incident face fluxes, two local input states, and local gravity. An adjacent cell may be updated independently using the shared immutable face flux. Abundance postprocessing adds only a cell-local species reduction, plus the specified region and diagnostic policy.

Output consistency and conservation. Floors may change mass and stored energy; abundance repair may change species partial masses. If the original density is below its floor, the scalar fractions before renormalization sum according to $\sum_a \bar{Z}_a / \rho^+$, rather than $\sum_a \bar{Z}_a / \bar{\rho}$. Likewise, magnetic energy is computed from the updated field before energy flooring. Replacing these operations with a different order would change the algorithm.

The literal source computes χ even in hydrodynamics. A new implementation may avoid this unused intermediate in that branch. The simplified damping exponential is algebraically equivalent on the normal nonzero-stage-scale path, but also gives a defined mathematical zero-stage-scale limit; that extension is not the behavior of the source's ungarded division. Floating-point regrouping can change rounding even when the mathematical dependency graph is preserved.

Source map and configuration correspondence

Mathematical item	Source item
V^0, V^s, V^+	solState_tmp, pre-update and overwritten hy_starState
q, e	ENER_VAR, EINT_VAR
$a, b, c, \Delta t, \Delta t_{\text{old}}$	coeffs(1:3), dt, dtOld
$\Delta x_d, \chi$	blockDesc%deltas, dhdt
$F_{C,d}^-, F_{C,d}^+$	Incident faces from hy_flx, hy_fly, hy_flz
$\mathbf{g}, \rho_{\min}, e_{\min}$	hy_grav, hy_smallldens, hy_smallE
α, c_h, p_B	hy_alphaGLM, hy_C_hyp, MAGP_VAR
$\phi^0, \phi^{\text{old}}, \phi^s$	Saved GPOT_VAR, saved GPOL_VAR, pre-update stage GPOT_VAR
x_g	gr_smallx in Grid_data

Routine	Arithmetic covered
hy_rk_updateSoln	Stage scale, direction spacings, cell-update region, and normalization call.
update_solution	Conservative construction; flux divergence and RK blend; GLM damping; magnetic energy; floors; recovered fractions; potential extrapolation.
gravSources	Stage density times gravity and momentum dotted with gravity; zero other sources.
Grid_renormAbundance	GridMain clipping, species sum, error test, normalization, and final clipping; alternatively stub.
Grid coordinate/area/volume access	Metadata for rejected non-Cartesian geometry; not additional active update formulas.

Branch coverage. Steps 1–3 and 5–6 describe Cartesian host and offloaded cell arithmetic. Gravity sources require `GRAVITY`; magnetic and cleaning formulas require `SPARK_GLM`; potential extrapolation requires `GPOT_VAR`. The abundance call is compiled when `NSPECIES+NMASS_SCALARS>0`. In a concrete grid build with zero species but positive other scalars, its species loops are empty and its computed sum is zero, so the current error test aborts; a stub build instead returns. No species-free normalization formula is added here to replace that existing behavior.

The update imports flux-correction settings and gravitational constants but does not use them in its active arithmetic. Flux correction, Poisson solving, EOS evaluation, and boundary updates belong to other call paths. The commented geometric-source expressions are excluded because the entry point rejects that geometry before updating any cells.